

**25** Which row correctly describes the adaptations of companion cells, phloem sieve tube elements and xylem vessel elements for their roles?

|          | companion cell                                         | phloem sieve tube element                              | xylem vessel element                                   |
|----------|--------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|
| <b>A</b> | a thin layer of cytoplasm to reduce resistance to flow | strong, lignified cell walls for supporting the plant  | many plasmodesmata for communication between cells     |
| <b>B</b> | empty cells allowing water to flow freely              | many plasmodesmata for communication between cells     | strong, lignified cell walls for supporting the plant  |
| <b>C</b> | many plasmodesmata for communication between cells     | a thin layer of cytoplasm to reduce resistance to flow | empty cells allowing water to flow freely              |
| <b>D</b> | strong, lignified cell walls for supporting the plant  | empty cells allowing water to flow freely              | a thin layer of cytoplasm to reduce resistance to flow |